REB, The Course
Class 15 Practice Assignment Solution

A jacketed, 50 gal BSTR is being considered for the conversion of A to Z, reaction (1). The reactor would be filled with a 60 °F solution containing reagent A at a concentration of 0.02 lbmol gal^{-1}. Saturated steam at 212 °F would condense in the jacket to provide heat. The conversion must reach 90% in 50 min.

The heat capacity of the reacting fluid is 1 BTU lb$_m^{-1}$, and its density is 8.5 lb$_m$ gal^{-1}. The heat of reaction (1) is –5150 BTU lbmol^{-1}. The rate coefficient exhibits Arrhenius temperature dependence with a pre-exponential factor of 4630 gal lbmol^{-1} min^{-1} and an activation energy of 6580 BTU lbmol^{-1}.

If the heat transfer coefficient is 37.3 BTU h^{-1} ft^{-2} °R^{-1}, what heat transfer area is needed, and what will be the final temperature of the reacting fluid?

$$A \rightarrow Z \tag{1}$$

$$r_1 = k_1 C_A^2 \tag{2}$$

Assignment Summary

Reaction

$$A \rightarrow Z \tag{1}$$

Rate Expression

$$r_1 = k_1 C_A^2 \tag{2}$$

Reactor: non-isothermal BSTR

Given Constants: V = 50 gal, T_0 = 60 °F, $C_{A,0}$ = 0.02 lbmol gal^{-1}, $C_{Z,0}$ = 0, T_{ex} = 212 °F, f_A = 90%, t_f = 50 min, $\tilde{C}_p$ = 1 BTU lb$_m^{-1}$, ρ = 8.5 lb$_m$ gal^{-1}, ΔH_1 = –5150 BTU lbmol^{-1}, $k_{0,1}$ = 4630 gal lbmol^{-1} min^{-1}, E_1 = 6580 BTU lbmol^{-1}, and U = 37.3 BTU h^{-1} ft^{-2} °R^{-1}.

Deliverables: A_{ex}, and T_f

Computational Strategy

This assignment involves a response task. There are no parameters that need to be varied, so the reactor design equations need to be solved once. That will be accomplished by calling the BSTR model function defined below. If the design equations contain a coupled unknown, it will be passed to the model function as an argument, and the coupled unknown residual function defined below will be used to calculate it. The deliverables will be generated by calling the deliverables function defined below which will coordinate the calculations. The constants listed above will be available within all of the functions defined below.

BSTR Model Function

BSTR Design Equations:

$$\frac{dn_A}{dt} = -r_1 V \tag{3}$$

$$\frac{dn_Z}{dt} = r_1 V \tag{4}$$

$$\frac{dT}{dt} = \frac{\dot{Q} - r_1 \Delta H_1 V}{\rho \tilde{C}_p V} \tag{5}$$

BSTR Model Variables: t, n_A, n_Z, and T

Additional Computable Unknowns: r_1, k_1, C_A, and $\dot{Q}$

$$k_1 = k_{0,1} \exp\left(\frac{-E_1}{RT}\right) \tag{6}$$

$$\dot{Q} = UA_{ex}\left(T_{ex} - T\right) \tag{7}$$

Coupled Unknown: A_{ex}

A_{ex} must be passed to the BSTR model function as an argument and it must satisfy implicit equation (8).

$$A_{ex} : n_A\Big|_{t_f} = n_{A,0}\left(1 - f_A\right) \quad \Rightarrow \quad 0 = n_A\Big|_{t_f} - n_{A,0}\left(1 - f_A\right) = \epsilon \qquad (8)$$

IVODE Solver Requirements:

1. Initial values

$$t\Big|_{t=0} = 0 \qquad (9)$$

$$n_i\Big|_{t=0} = C_{i,0}V \qquad i = A \text{ and } Z \qquad (10)$$

$$T\Big|_{t=0} = T_0 \qquad (11)$$

2. Stopping criterion

$$t = t_f \qquad (12)$$

3. Derivatives function that
   a. receives the BSTR model variables (t, n_A, n_Z, and T) at the start of an integration step,
   b. calculates the additional computable unknowns using equations (2), (6), and (7), and
   c. evaluates and returns the BSTR design equation derivatives using equations (3), (4), and (5).

Coupled Unknown Residual Function

1. Receives a guess for A_{ex} as an argument
2. Calls the BSTR function using the guess as the argument to get $\underline{t}$, $\underline{n}_A$, $\underline{n}_Z$, and $\underline{T}$
3. Evaluates and returns the coupled unknown residual using equation (8)

Deliverables Function

1. Defines an initial guess for A_{ex}

2. Calls an ATE solver passing the initial guess and the coupled unknown residual function as arguments to get A_{ex}
3. Calls the BSTR function using A_{ex} as the argument to get $\underline{t}$, $\underline{n}_A$, $\underline{n}_Z$, and $\underline{T}$
4. Extracts the final temperature

$$T_f = \underline{T}\Big|_{t=t_f} \tag{13}$$

Implementation of the Calculations

The Python script, practice_15.py, and Matlab function, practice_15.m,that were written to perform the calculations accompany this solution.

Results and Discussion

A heat transfer area of 0.44 ft² is required to achieve 90% conversion in 50 min. The final temperature is 75.4 °F. The temperature increase is only 15.4 °F, and it occurs over a period of 50 minutes, so it is unlikely that there is a thermal runaway. Nonetheless, the temperature profile, Figure 1, can be plotted to check for a sharp temperature increase. The figure shows the temperature rise is acceptable.

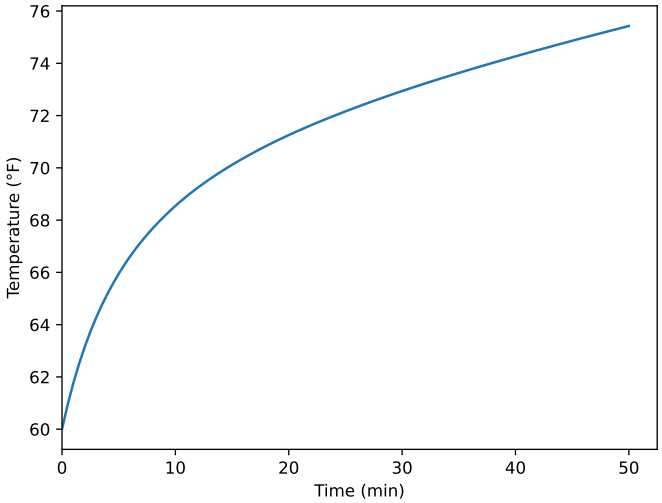


Figure 1. Temperature profile during processing in a BSTR with a heat transfer area of 0.44 ft².

Summary

To achieve 90% conversion in 50 min, the heat transfer area must be 0.44 ft^2, and the final temperature will be 75.4 °F.